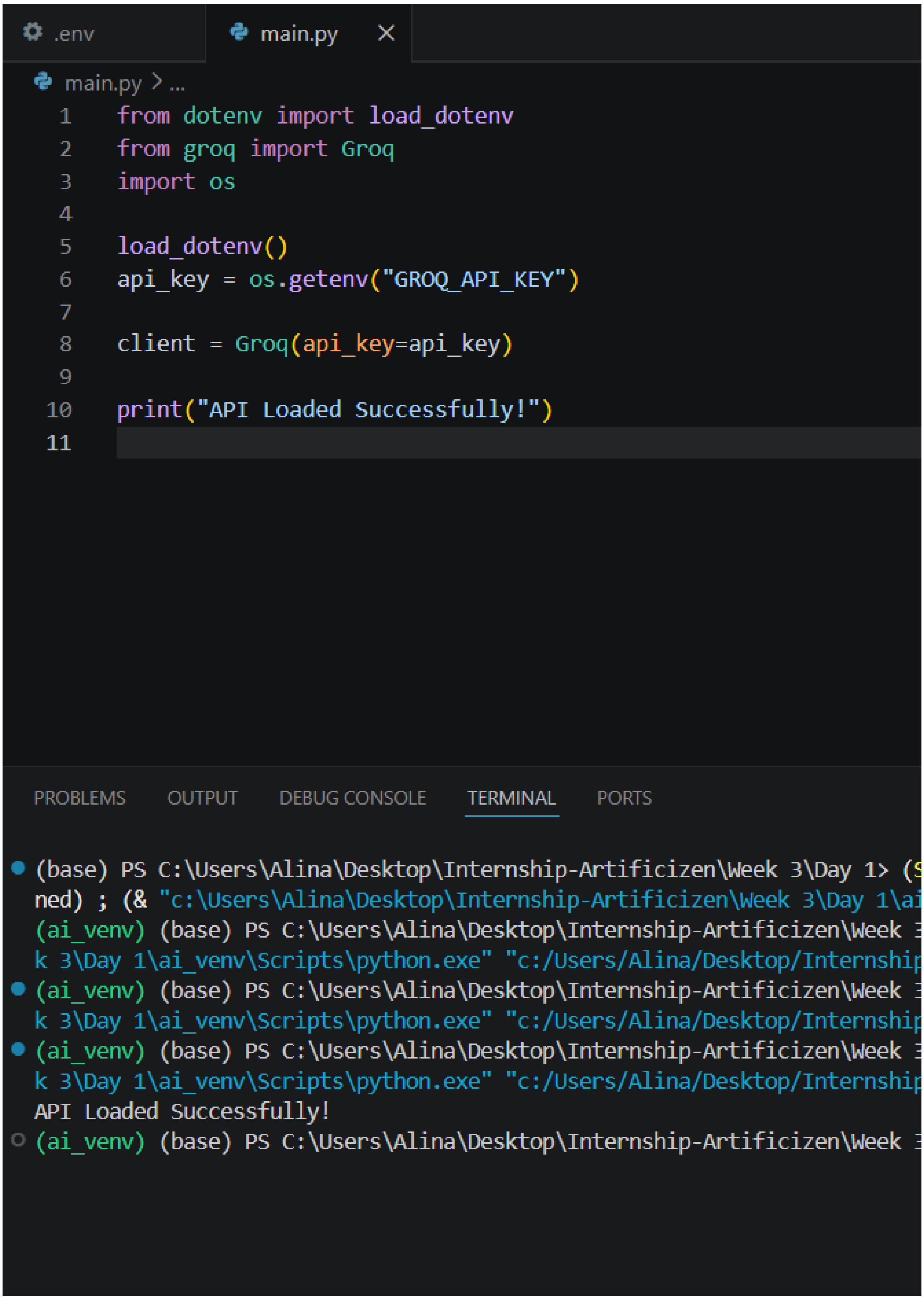
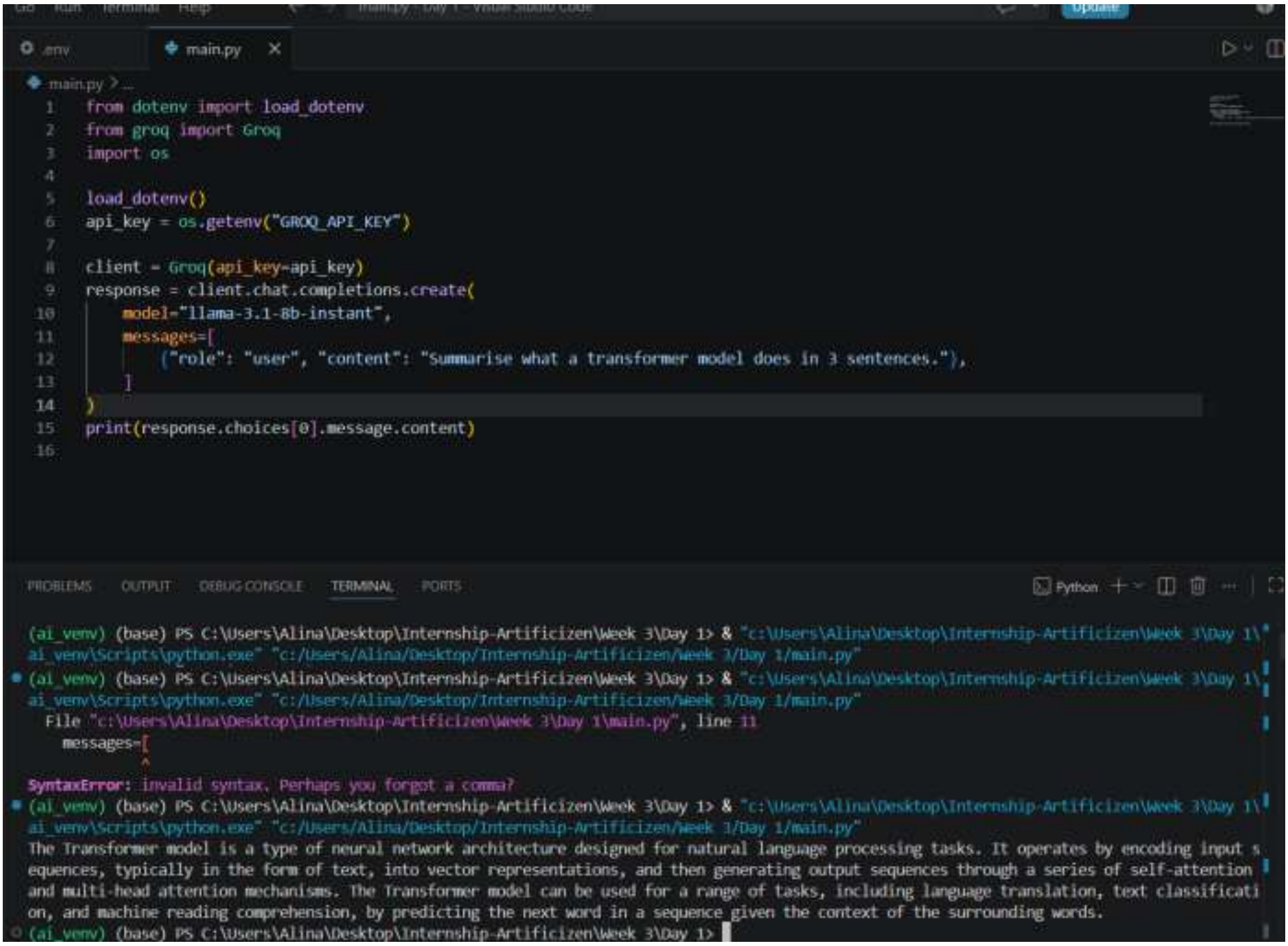
Practice Questions

1. Sign up at console.groq.com, generate an API key, store it in .env as GROQ_API_KEY, and load it with python-dotenv. Never hardcode it.



2. Call the Groq API using llama-3.1-8b-instant and print a completion for: “Summarise what a transformer model does in 3 sentences.”



3. Run the same prompt three times at temperature=0 and three times at temperature=1.0. Print all six responses and write a one-line observation about the difference.

Temperature = 0:

1st:

```python
from groq import Groq
import os

load_dotenv()
api_key = os.getenv("GROQ_API_KEY")

client = Groq(api_key=api_key)
response = client.chat.completions.create(
    model="llama-3.1-8b-instant",
    messages=[
        {"role": "system", "content": "You are an AI Tutor."},
        {"role": "user", "content": "Summarise what a transformer model does in 3 sentences."},
    ],
    temperature=0,
)
print(response.choices[0].message.content)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
The Transformer model is a type of neural network architecture designed for natural language processing tasks. It operates by encoding input s
equences, typically in the form of text, into vector representations, and then generating output sequences through a series of self-attention
and multi-head attention mechanisms. The Transformer model can be used for a range of tasks, including language translation, text classificati
on, and machine reading comprehension, by predicting the next word in a sequence given the context of the surrounding words.
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
slation, text summarization, and sentiment analysis. It works by breaking down input sequences into smaller chunks called tokens, and then use
s self-attention mechanisms to weigh the importance of each token in relation to the others, allowing the model to capture complex relationshi
ps and dependencies within the input data. This enables the transformer model to generate highly accurate and contextually relevant outputs, m
aking it a powerful tool for a wide range of NLP applications.
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1>
```

2nd:

```
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
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aking it a powerful tool for a wide range of NLP applications.
```

3rd:

The responses at t=0 are identical

```
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
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s self-attention mechanisms to weigh the importance of each token in relation to the others, allowing the model to capture complex relationshi
ps and dependencies within the input data. This enables the transformer model to generate highly accurate and contextually relevant outputs, m
aking it a powerful tool for a wide range of NLP applications.
```

At Temperature: 1.0:

The responses have different wording, examples, or sentence structure, while still answering the same question.

```python
load_dotenv()
api_key = os.getenv("GROQ_API_KEY")

client = Groq(api_key=api_key)
response = client.chat.completions.create(
    model="llama-3.1-8b-instant",
    messages=[
        {"role": "system", "content": "You are an AI Tutor."},
        {"role": "user", "content": "Summarise what a transformer model does in 3 sentences."},
    ],
    temperature=1.0,
)
print(response.choices[0].message.content)
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
The Transformer model is a type of deep learning architecture used primarily for tasks such as machine translation, text classification, and n
atural language processing. It works by breaking down input text into smaller segments, known as tokens, and then uses self-attention mechanis
ms to weigh the importance of each token when generating an output. This process is done without the need for recurrent neural networks, allow
ing the model to process large sequences of text much more efficiently and in parallel.
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
A transformer model is a type of deep learning architecture primarily used for natural language processing tasks. It processes input sequences
, such as text, through self-attention mechanisms that allow it to weigh the relevance of different words or tokens in relation to one another
, enabling the model to effectively capture contextual relationships. By using this self-attendance and layer-based approach, the transformer
model can generate a wide range of outputs, from machine translation to text generation and more.
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
The Transformer model is a type of deep learning architecture used in natural language processing tasks, such as language translation, text su
mmarization, and question answering. It processes sequential data by using self-attention mechanisms to weigh the importance of different inpu
t elements and generate representations for each position in the sequence, allowing it to efficiently capture long-range dependencies and cont
ext. Essentially, the Transformer model can analyze and generate human-like language by leveraging these attention-based representations and l
everaging parallel processing to improve computational efficiency.
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1>
```

Simple and easy way to do this is:

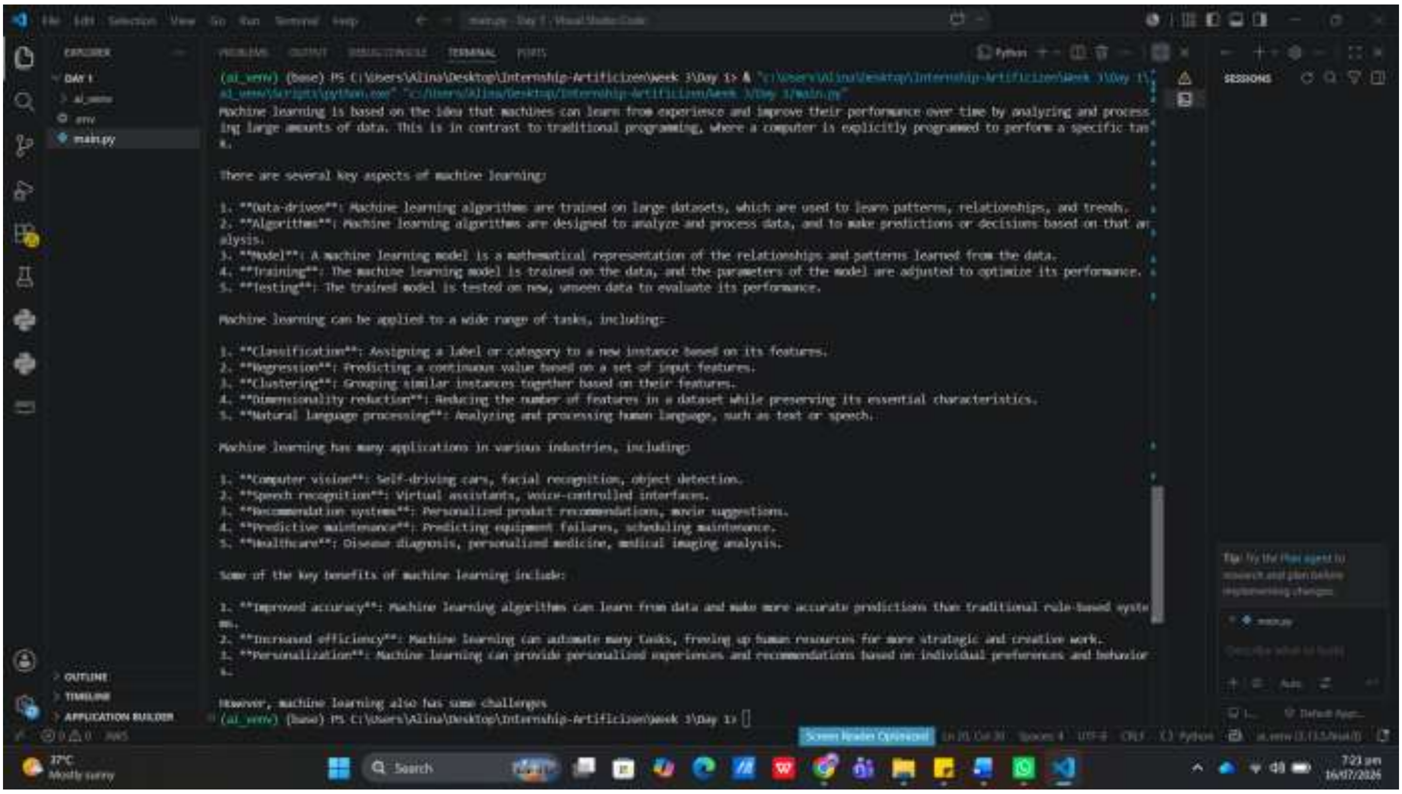
Using loop:

```
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\
ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
Temperature = 0

Run 1:
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
slation, text summarization, and sentiment analysis. It works by breaking down input sequences into smaller chunks, analyzing the relationship
s between these chunks, and then reassembling the information to generate the desired output. The transformer model uses self-attention mechan
isms to weigh the importance of different input elements, allowing it to capture long-range dependencies and contextual relationships within t
he input data.

Run 2:
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
slation, text summarization, and sentiment analysis. It works by breaking down input sequences into smaller chunks, analyzing the relationship
s between these chunks, and then reassembling the information to generate the desired output. The transformer model uses self-attention mechan
isms to weigh the importance of different input elements, allowing it to capture long-range dependencies and contextual relationships within t
he input data.

Run 3:
A transformer model is a type of neural network architecture primarily used for natural language processing (NLP) tasks, such as language tran
slation, text summarization, and sentiment analysis. It works by breaking down input sequences into smaller chunks, analyzing the relationship
s between these chunks, and then reassembling the information to generate the desired output. The transformer model uses self-attention mechan
isms to weigh the importance of different input elements, allowing it to capture long-range dependencies and contextual relationships within t
he input data.
```

Code:

```python
client = Groq(api_key=os.getenv("GROQ_API_KEY"))

prompt = "Summarise what a transformer model does in 3 sentences."

print("Temperature = 0\n")

for i in range(3):
    response = client.chat.completions.create(
        model="llama-3.1-8b-instant",
        messages=[
            {
                "role": "user",
                "content": prompt
            }
        ],
        temperature=0
    )

    print(f"Run {i+1}:")
    print(response.choices[0].message.content)
    print()

print("Temperature = 1.0\n")
```

4. Write a wrapper function ask(prompt, system=None, model='llama-3.1-8b-instant', temperature=0.7, max_tokens=512) that calls Groq and returns only the text string. This function will be reused every day this week.



Code:

```python
from dotenv import load_dotenv
from groq import Groq
import os

load_dotenv()
api_key = os.getenv("GROQ_API_KEY")

client = Groq(api_key=api_key)
def ask(prompt,system=None,model="llama-3.1-8b-instant",temperature=0.7, max_tokens=512):
    messages = []
    if system:
        messages.append({"role":"system", "content":system})
    if prompt:
        messages.append({"role":"user","content":prompt})

    response=client.chat.completions.create(
        model=model,
        messages=messages,
        temperature=temperature,
        max_tokens=max_tokens
    )
    return response.choices[0].message.content

answer=ask("What is Machine Learning?")
print(answer)
```

5. Pass a system message: "You are a strict JSON-only responder. Never output anything outside a JSON object." Ask any question and print the raw output. Did it obey?

Yes, it did obey the system instruction

```python
client = Groq(api_key=api_key)
def ask(prompt,system,model="llama-3.1-8b-instant",temperature=0.7, max_tokens=512):
    messages = []
    if system:
        messages.append({"role":"system", "content":system})
    if prompt:
        messages.append({"role":"user","content":prompt})

    response=client.chat.completions.create(
        model=model,
        messages=messages,
        temperature=temperature,
        max_tokens=max_tokens
    )
    return response.choices[0].message.content

answer=ask("What is Machine Learning?","You are a strict JSON-only responder. Never output anything outside a JSON objec
print(answer)
```

```
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
{
  "Machine Learning": {
    "definition": "A field of study that focuses on the use of algorithms and statistical models to enable machines to perform a specific task without using explicit instructions, relying on patterns and data to make decisions.",
    "subfields": ["Supervised Learning", "Unsupervised Learning", "Reinforcement Learning"],
    "key_features": ["Pattern recognition", "Data analysis", "Model training", "Model evaluation", "Model deployment"]
  }
}
```

6. Call the API with llama-3.3-70b-versatile and llama-3.1-8b-instant for the same prompt. Print both responses and the usage.total_tokens for each. Note the difference in response quality and latency.

```python
def ask(prompt,system,model="llama-3.1-8b-instant",temperature=0.7, max_tokens=512):

    response=client.chat.completions.create(
        model=model,
        messages=messages,
        temperature=temperature,
        max_tokens=max_tokens
    )
    return response
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(ai_venv) (base) PS C:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1> & "c:\Users\Alina\Desktop\Internship-Artificizen\Week 3\Day 1\ai_venv\Scripts\python.exe" "c:/Users/Alina/Desktop/Internship-Artificizen/Week 3/Day 1/main.py"
{
  "Machine Learning": {
    "definition": "A subfield of artificial intelligence that involves the use of algorithms and statistical models to enable computers to learn from data and make predictions or decisions without being explicitly programmed.",
    "key_characteristics": ["Pattern recognition", "Data analysis", "Algorithmic learning"],
    "types": ["Supervised learning", "Unsupervised learning", "Reinforcement learning"]
  }
}

Prompt Tokens: 56
Completion Tokens: 88
Total Tokens: 144
Latency: 0.98 seconds

Model: llama-3.3-70b-versatile
{"response": "Machine Learning is a subset of Artificial Intelligence that involves training algorithms to learn patterns and make predictions or decisions based on data, without being explicitly programmed."}

Prompt Tokens: 56
Completion Tokens: 35
Total Tokens: 91
Latency: 0.27 seconds
```